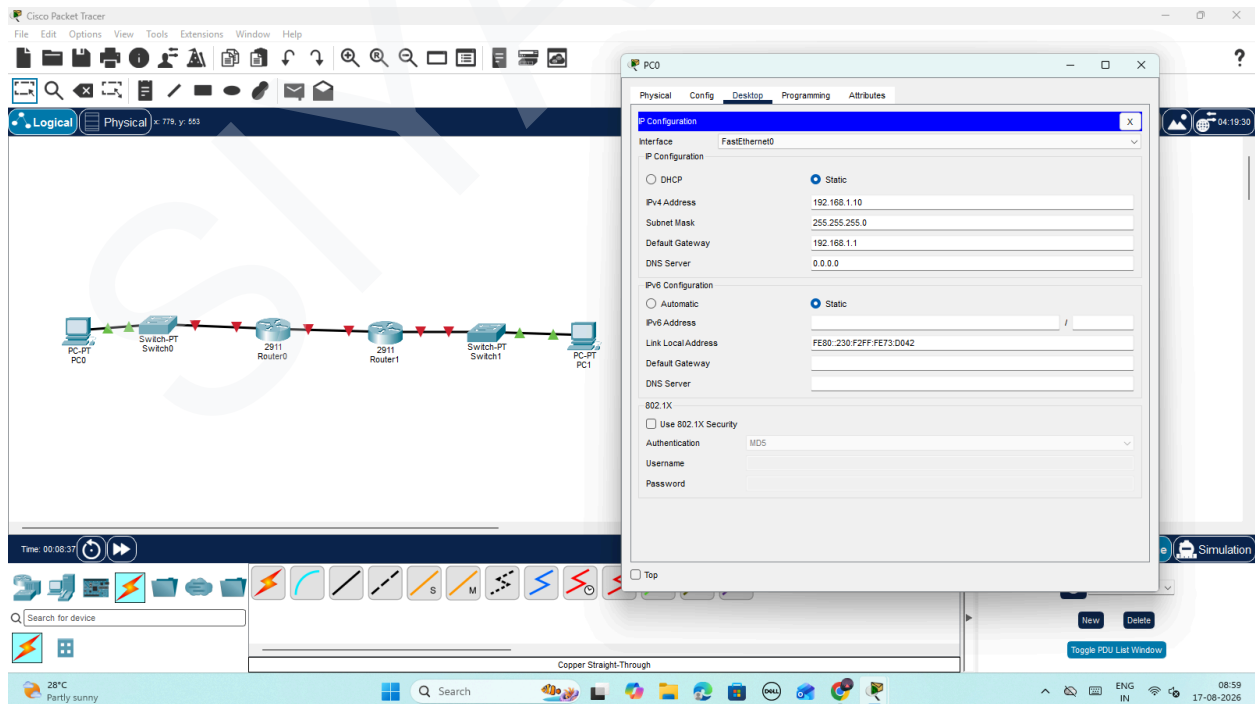
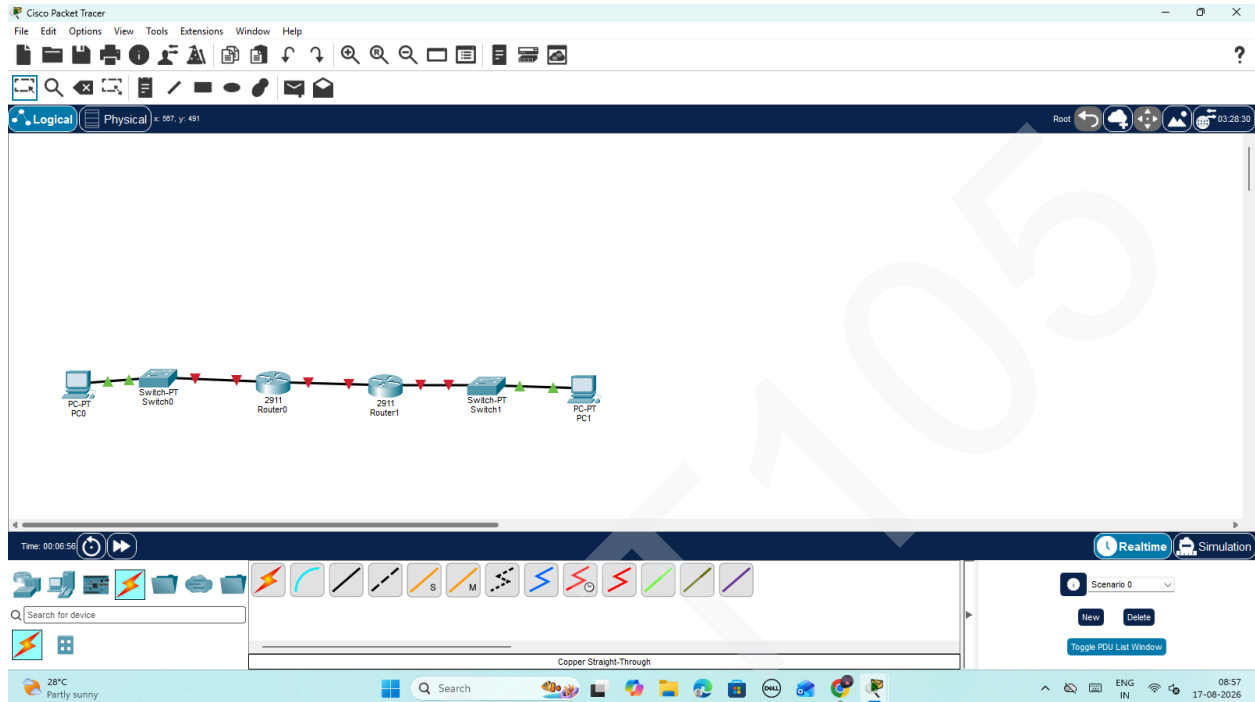


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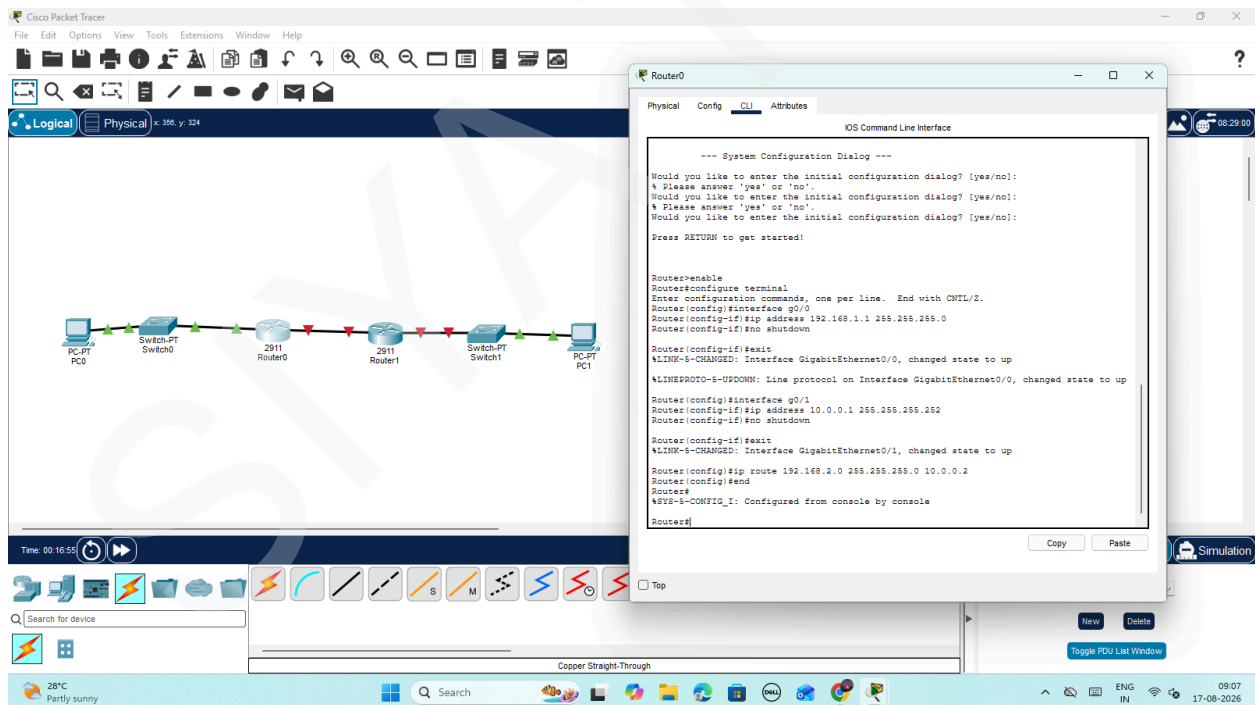
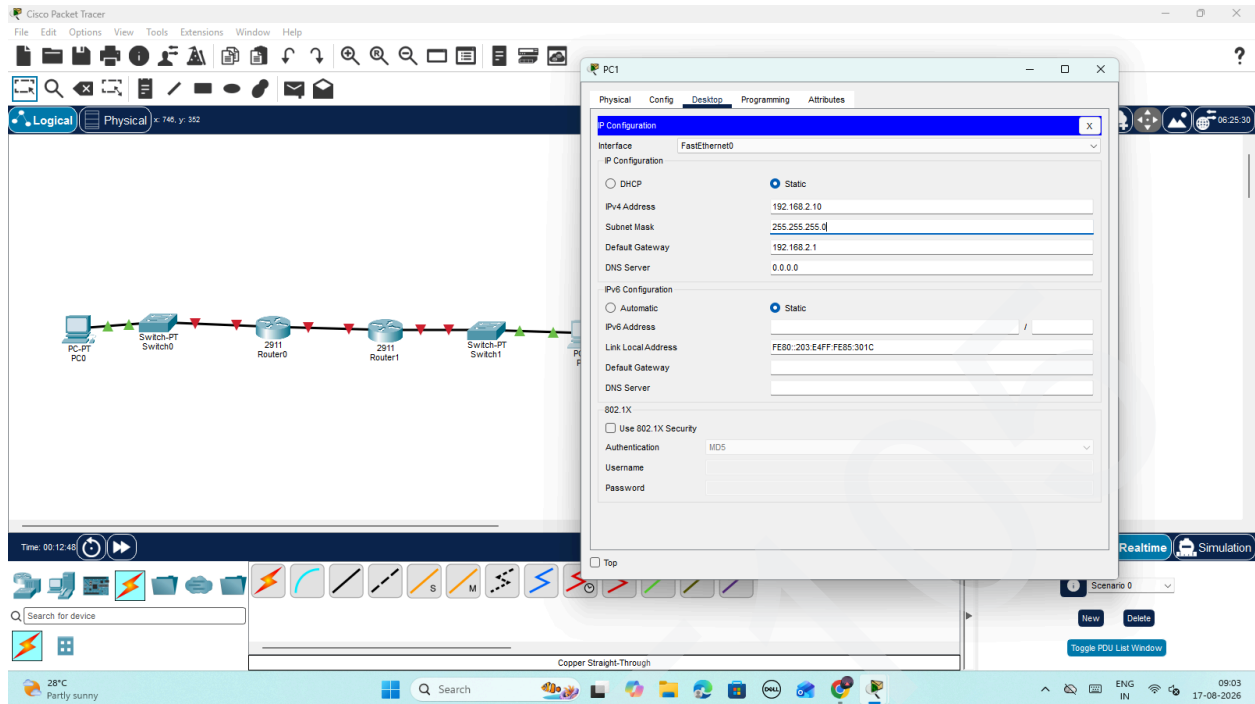
AIM: Configure IPsec on network devices to provide secure communication and protect against unauthorized access and attacks.



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The image displays two screenshots of the Cisco Packet Tracer software interface, showing network configurations and command-line interfaces for two routers, Router1 and Router0.

Top Screenshot (Router1):

- The main window shows a network topology with two PCs (PC0 and PC1) connected to two switches (Switch0 and Switch1), which are connected to two routers (Router0 and Router1).
- The CLI window for Router1 shows the following configuration commands and output:

```
Router1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router1(config)#interface g0/0
Router1(config-if)#ip address 192.168.2.1 255.255.255.0
Router1(config-if)#no shutdown
Router1(config-if)#exit
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
Router1(config)#interface g0/1
Router1(config-if)#ip address 10.0.0.2 255.255.255.252
Router1(config-if)#no shutdown
Router1(config-if)#exit
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
Router1(config)#ip route 192.168.1.0 255.255.255.0 10.0.0.1
Router1(config)#end
Router1#
SYS-5-CONFIG_I: Configured from console by console

Router1#show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 192.168.2.1 YES manual up up
GigabitEthernet0/1 10.0.0.2 YES manual up up
GigabitEthernet0/2 unassigned YES unset administratively down down
Vlan1 unassigned YES unset administratively down down
Router1#
```

Bottom Screenshot (Router0):

- The main window shows a network topology with two PCs (PC0 and PC1) connected to two switches (Switch0 and Switch1), which are connected to two routers (Router0 and Router1).
- The CLI window for Router0 shows the following configuration commands and output:

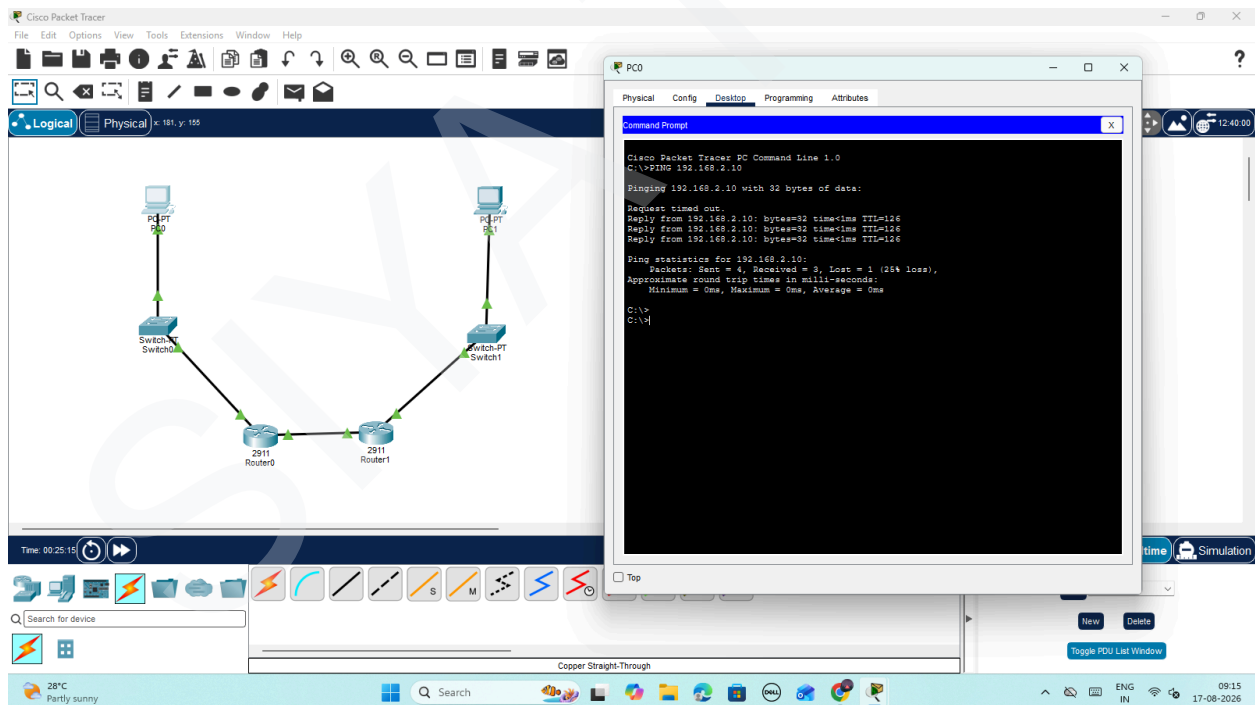
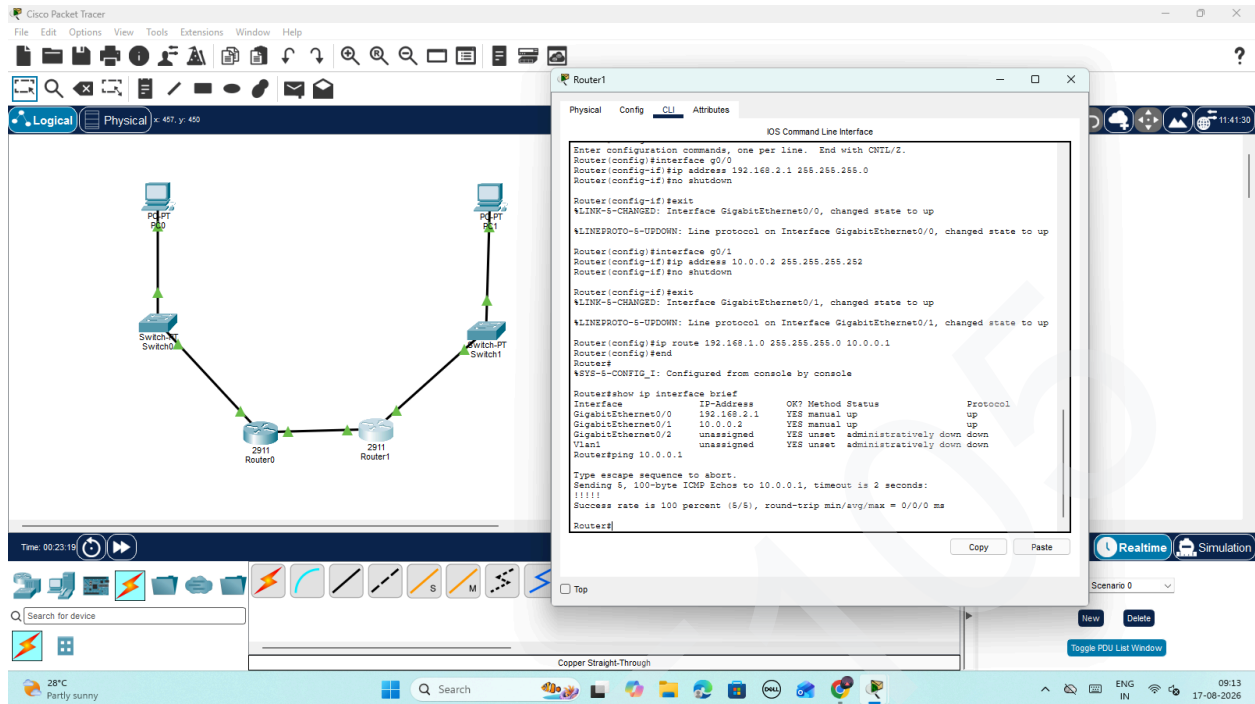
```
Router0#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router0(config)#interface g0/0
Router0(config-if)#ip address 192.168.1.1 255.255.255.0
Router0(config-if)#no shutdown
Router0(config-if)#exit
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
Router0(config)#interface g0/1
Router0(config-if)#ip address 10.0.0.1 255.255.255.252
Router0(config-if)#no shutdown
Router0(config-if)#exit
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
Router0(config)#ip route 192.168.2.0 255.255.255.0 10.0.0.2
Router0(config)#end
Router0#
SYS-5-CONFIG_I: Configured from console by console

Router0#
Router0#
Router0#
Router0#ping 10.0.0.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echoes to 10.0.0.2, timeout is 2 seconds:
!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0 ms
Router0#
```

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The image displays two screenshots of the Cisco Packet Tracer interface, showing a network configuration for IPsec.

Top Screenshot: The network topology shows two routers (2911) connected via switches and PCs. The configuration window for Router0 is open, showing the initial setup steps:

```
Router0>configure terminal
Router0(config)#crypto isakmp policy 10
Router0(config-isakmp)#
Router0#
$SYS-6-CONFIG_I: Configured from console by console
```

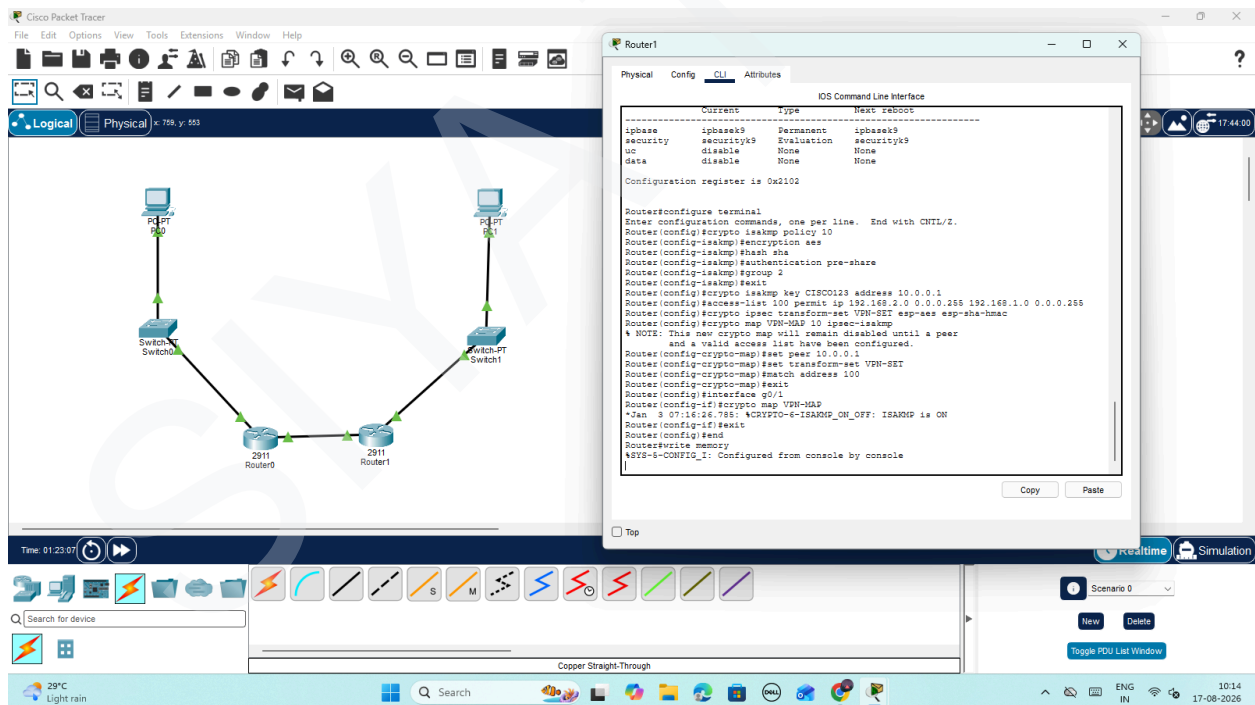
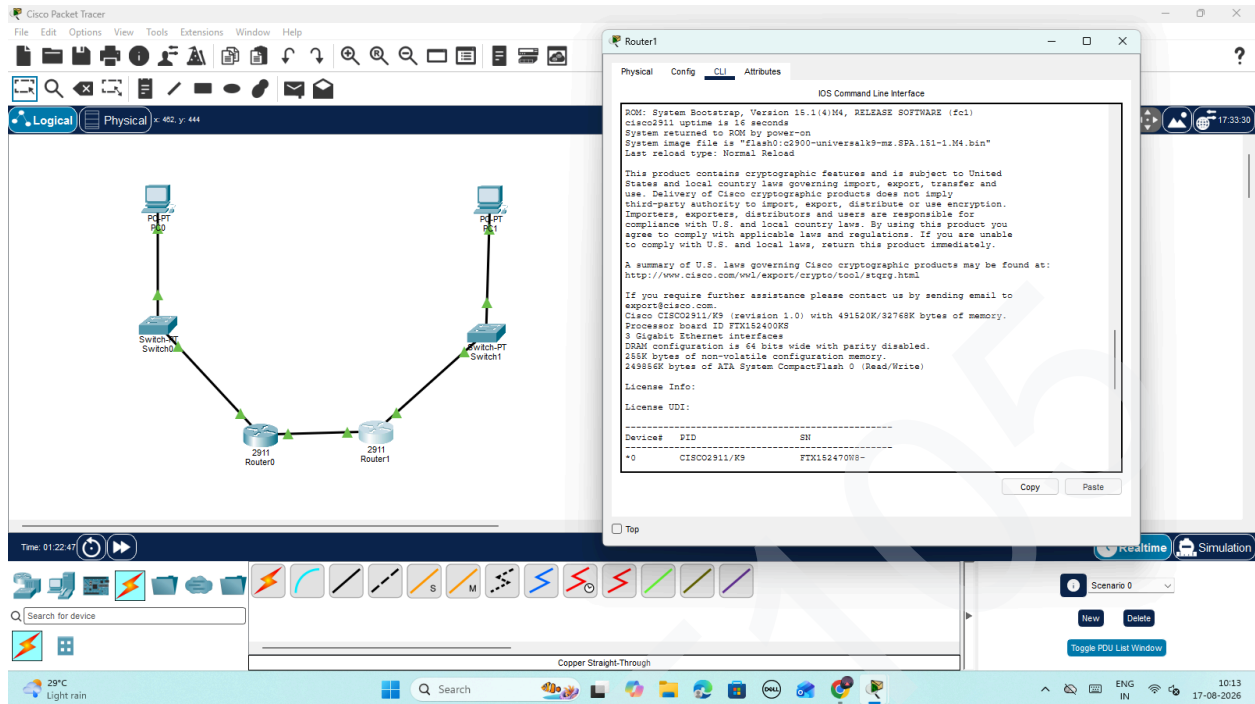
Bottom Screenshot: The configuration window for Router0 is open, showing the continuation of the IPsec configuration:

```
$SYS-6-CONFIG_I: Configured from console by console
Building configuration...
[OK]
Router0>configure terminal
Router0(config)#crypto isakmp policy 10
Router0(config-isakmp)#encryption aes
Router0(config-isakmp)#auth sha
Router0(config-isakmp)#authentication pre-share
Router0(config-isakmp)#group 2
Router0(config-isakmp)#exit
Router0(config)#crypto isakmp key CISC0123 address 10.0.0.2
Router0(config)#access-list 100 permit ip 192.168.1.0 0.0.0.255 192.168.2.0 0.0.0.255
Router0(config)#crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
Router0(config-crypto-map)#set peer 10.0.0.2
Router0(config-crypto-map)#set transform-set VPN-SET
Router0(config-crypto-map)#match address 100
Router0(config-crypto-map)#exit
Router0(config)#interface g0/1
Router0(config-if)#crypto map VPN-MAP
*Jan 3 07:16:24.788: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
Router0(config-if)#exit
Router0(config)#interface g0/1
Router0(config-if)#crypto map VPN-MAP
*Jan 3 07:16:24.788: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
Router0(config-if)#exit
Router0(config)#end
Router0#write memory
$SYS-6-CONFIG_I: Configured from console by console
Building configuration...
[OK]
Router0#
```

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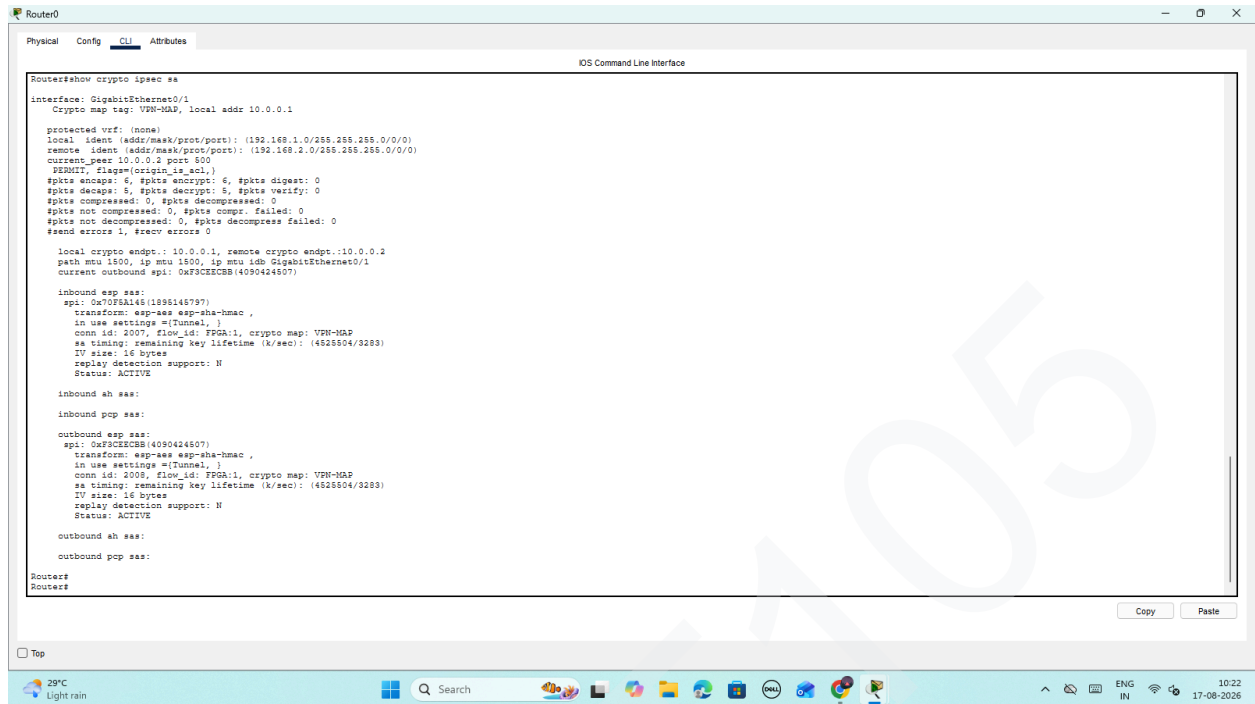
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```
Router0#show crypto ipsec sa
interface: GigabitEthernet0/1
Crypto map tag: VPN-MAP, local addr 10.0.0.1

protected vrf: (none)
local ident (addr/mask/prot/port): (192.168.1.0/255.255.255.0/0/0)
remote ident (addr/mask/prot/port): (192.168.2.0/255.255.255.0/0/0)
current_peer 10.0.0.2 port 500
  PERMIT, flags=(origin_is_acl,):
    #pkts encaps: 6, #pkts encrypt: 6, #pkts digest: 0
    #pkts decaps: 5, #pkts decrypt: 5, #pkts verify: 0
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 1, #recv errors 0

local crypto endpt.: 10.0.0.1, remote crypto endpt.: 10.0.0.2
path mtu 1500, ip mtu 1500, ip mtu idb GigabitEthernet0/1
current outbound spi: 0x79C8E8CB(4090424507)

inbound esp sas:
  spi: 0x70F8A146(1895146797)
  transform: esp-aes esp-sha-hmac ,
  in use settings = (Tunnel, )
  conn id: 2007, flow_id: FPOA:1, crypto map: VPN-MAP
  sa timing: remaining key lifetime (k/sec): (4525504/3283)
  IV size: 16 bytes
  replay detection support: N
  Status: ACTIVE

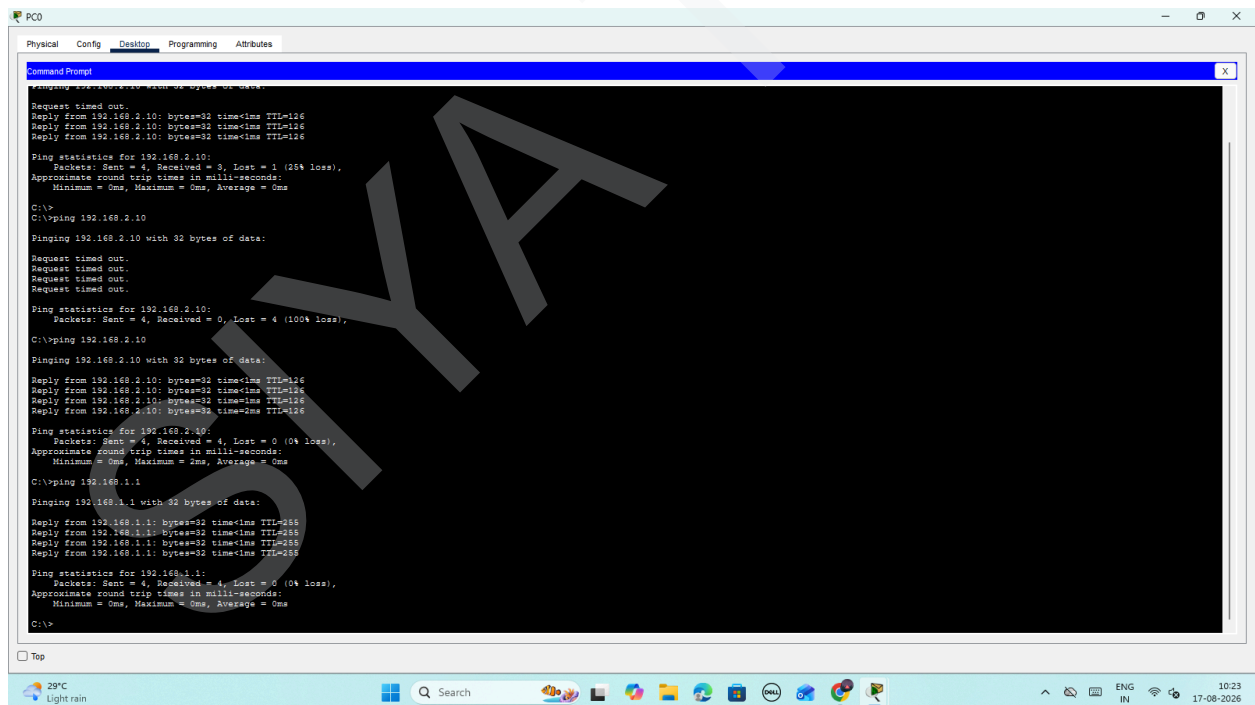
inbound ah sas:

inbound pop sas:

outbound esp sas:
  spi: 0x79C8E8CB(4090424507)
  transform: esp-aes esp-sha-hmac ,
  in use settings = (Tunnel, )
  conn id: 2008, flow_id: FPOA:1, crypto map: VPN-MAP
  sa timing: remaining key lifetime (k/sec): (4525504/3283)
  IV size: 16 bytes
  replay detection support: N
  Status: ACTIVE

outbound ah sas:

outbound pop sas:
```



```
C:\>ping 192.168.2.10
Pinging 192.168.2.10 with 32 bytes of data:
Request timed out.
Reply from 192.168.2.10: bytes=32 time=1ms TTL=126
Reply from 192.168.2.10: bytes=32 time=1ms TTL=126
Reply from 192.168.2.10: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
C:\>ping 192.168.2.10
Pinging 192.168.2.10 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.2.10
Pinging 192.168.2.10 with 32 bytes of data:
Reply from 192.168.2.10: bytes=32 time=1ms TTL=126
Reply from 192.168.2.10: bytes=32 time=1ms TTL=126
Reply from 192.168.2.10: bytes=32 time=1ms TTL=126
Reply from 192.168.2.10: bytes=32 time=2ms TTL=126

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>ping 192.168.1.1
Pinging 192.168.1.1 with 32 bytes of data:
Reply from 192.168.1.1: bytes=32 time=1ms TTL=255
Reply from 192.168.1.1: bytes=32 time=1ms TTL=255
Reply from 192.168.1.1: bytes=32 time=1ms TTL=255
Reply from 192.168.1.1: bytes=32 time=1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```